

SpatialExperiment and SpaceTrooper

Unified infrastructure and quality control workflows
for spatial omics data

Bioconductor workshop

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Who is Dario Righelli

Computer science background, bioinformatics practice, Bioconductor infrastructure.

Academic role

- Assistant Professor RTT
- Molecular Biology (DiBio)
- University of Padova
- Teaching:
Python, Statistics,
Machine Learning,
Bioinformatics

Research

- Single-cell and spatial transcriptomics
- Multimodal data integration
- Graph-based models
- Immunogenicity and pMHC modeling

Bioconductor

- Package development and infrastructure
- SpatialExperiment co-author
- SpaceTrooper maintainer
- Community working groups



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**Workshop perspective:
common infrastructure first,
specialization second.**

What is Bioconductor?

An open community open-source ecosystem where data structures and methods are designed to interoperate.

Infrastructure

- Shared experiment classes
- S4 conventions
- Assays + metadata + coordinates
- Standard accessors

Methods

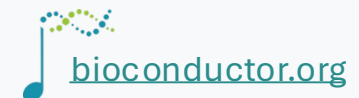
- Packages can operate on common representations instead of vendor-specific file layouts.
- Packages have to provide vignettes to help the user on the usage.

Open Community

- Open development
- Reproducible releases
- Vignettes, books and workflows
- Interoperability across packages
- Join it via:
 - Zulip: <https://tinyurl.com/bioczuliplink>
 - Forum: <https://support.bioconductor.org/>
 - Website: <https://bioconductor.org/>

For this workshop, the important design principle is:

Represent the experiment once → Reuse methods many times



Bioconductor books: workflows as executable documentation

A collection of community-coordinated efforts presented in book format, combining key challenges with computational approaches to address them.

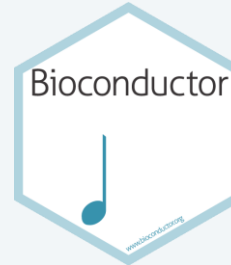
 **Bioconductor Books**

A selection of Bioconductor Online Books for learning Bioconductor, contributed by members of the Bioconductor community.

Modern Statistics for Modern Biology	→
Orchestrating Single-Cell Analysis with Bioconductor	→
Orchestrating Hi-C Analysis with Bioconductor	→
Orchestrating Spatial Transcriptomics Analysis with Bioconductor	→
R for Mass Spectrometry	→
SingleRBook	→
csawBook	→
BiocBookDemo	→
BiocPy	→
Orchestrating Microbiome Analysis with Bioconductor	→
scrapbook	→

OSCA

- Orchestrating Single-Cell Analysis with Bioconductor
- A reference for single-cell infrastructure and analysis patterns.



OSTA

- Orchestrating Spatial Transcriptomics Analysis with Bioconductor
- Covers infrastructure, import, QC, preprocessing, visualization and downstream analysis.



<https://www.bioconductor.org/help/bioconductor-books/>

In this workshop:

OSTA gives us the map; SpatialExperiment gives us the common representation.

A workshop in three acts

Concepts in the slides; code and interpretation in the Quarto.

1 • Infrastructure

- SpatialExperiment
- Synthetic spot + cell examples
- Real Visium + seqFISH
- Molecule-level assays
- IO ecosystem



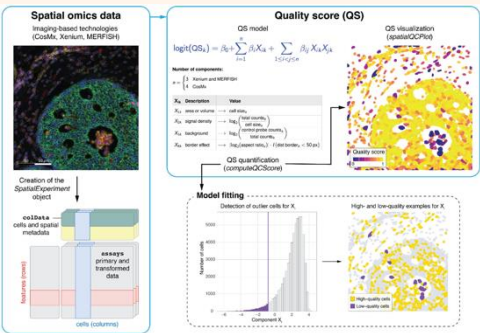
2 • Spatial QC

- CosMx mouse brain
- SpaceTrooper metrics
- FOV structure and border effects
- Data-driven QScore



3 • Interpretation

- Map quality back to tissue
- Inspect suspicious spatial regions
- Thresholds as decisions
- Self-paced exercise



The slides frame the story. The Quarto carries the hands-on.

Spatial omics is not one data type

Different technologies generate different spatial units and metadata.

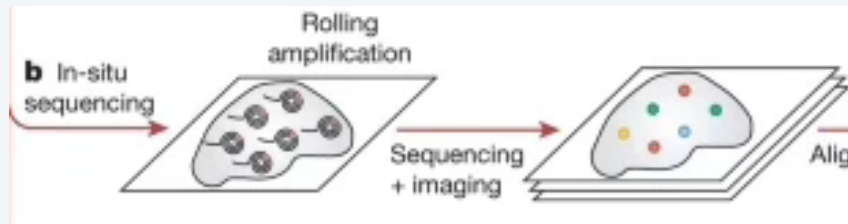
Sequencing-based

Typical unit: spot or bin

Examples

- Visium / Visium HD
- Slide-seq

Expression is aggregated over a capture location.



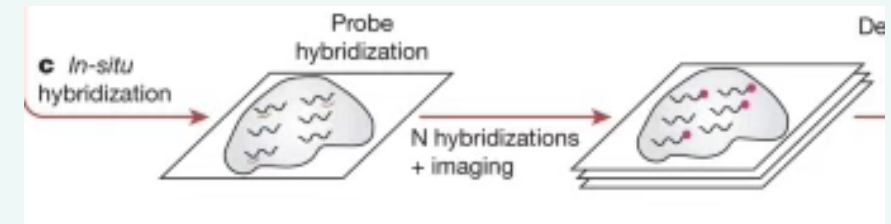
Imaging-based

Typical unit: segmented cell

Examples

- Xenium
- CosMx
- MERSCOPE

Can retain centroids, boundaries and individual molecules.

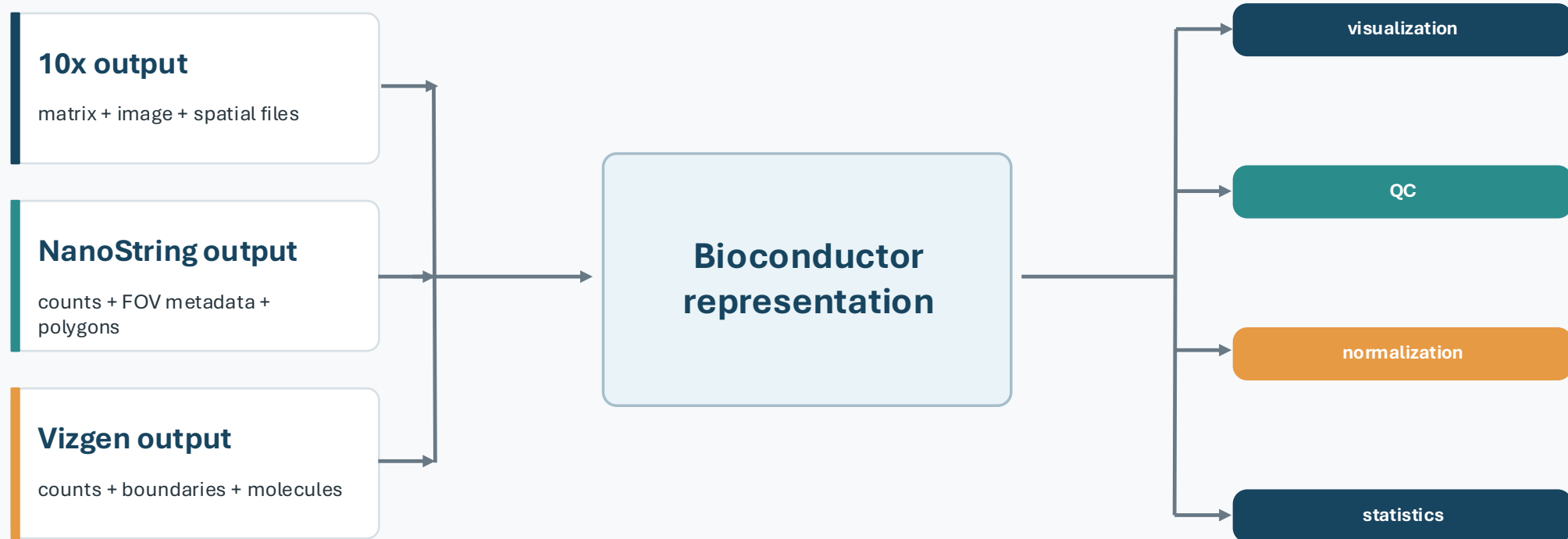


Different spatial information → common interfaces

Images adapted from Rao A. et al, Nature, 2021

The infrastructure problem

Downstream methods should not need to understand every vendor file layout.



Reading files and analyzing experiments are different problems.

SpatialExperiment: common representation infrastructure

Built on the SingleCellExperiment / SummarizedExperiment infrastructure.



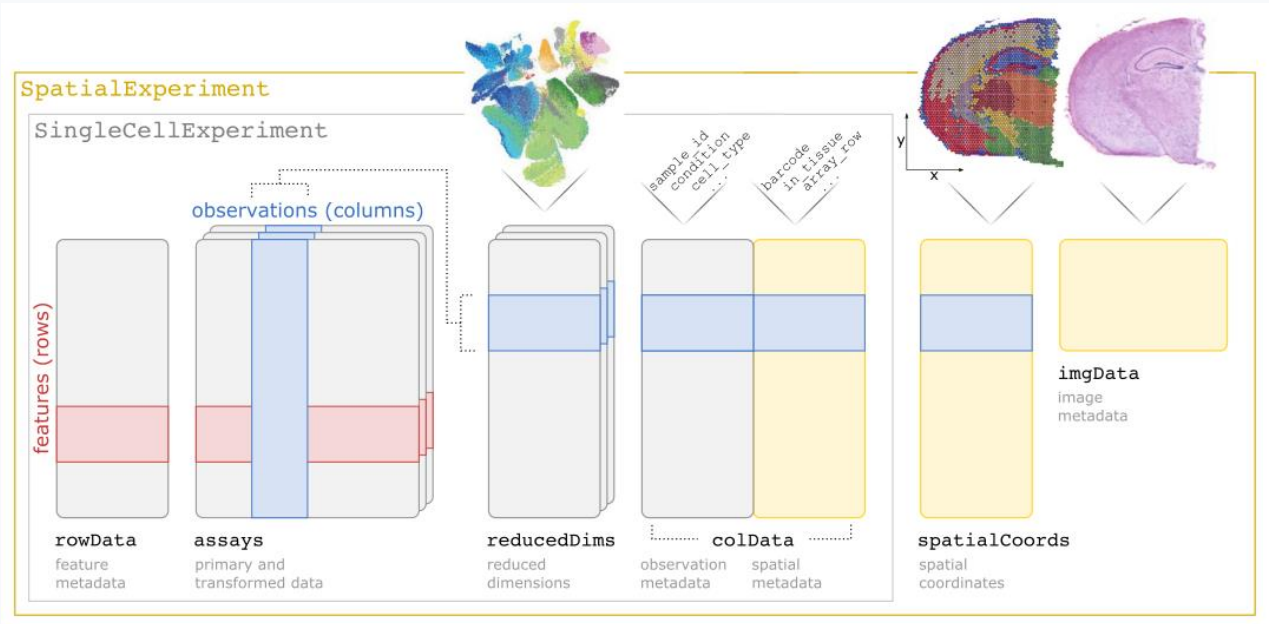
SummarizedExperiment



SingleCellExperiment



SpatialExperiment



The observational unit can change;
the interface does not.

SpatialExperiment describes what the experiment contains
adding accessors to images, cells and molecules coordinates

Righelli, Crowell, Weber, et al, Bioinformatics, 2021

One infrastructure, different spatial units

Hands-on: spots → cells → individual molecules.

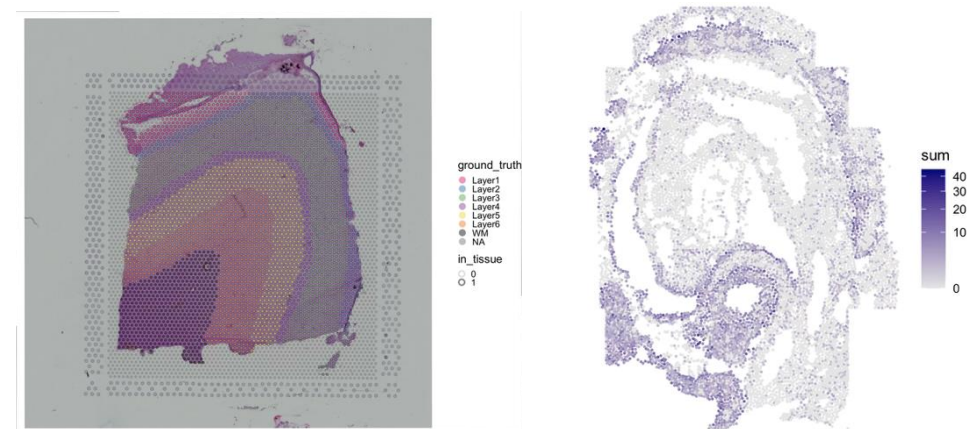
Sequencing-based

- Observation = spot / bin
- Counts matrix
- Spatial coordinates
- Optional tissue image
- Real example: Visium

Imaging-based

- Observation = segmented cell
- Counts matrix
- Cell centroids
- Boundaries / polygons
- Real example: seqFISH

Molecule-level information can still live inside the experiment



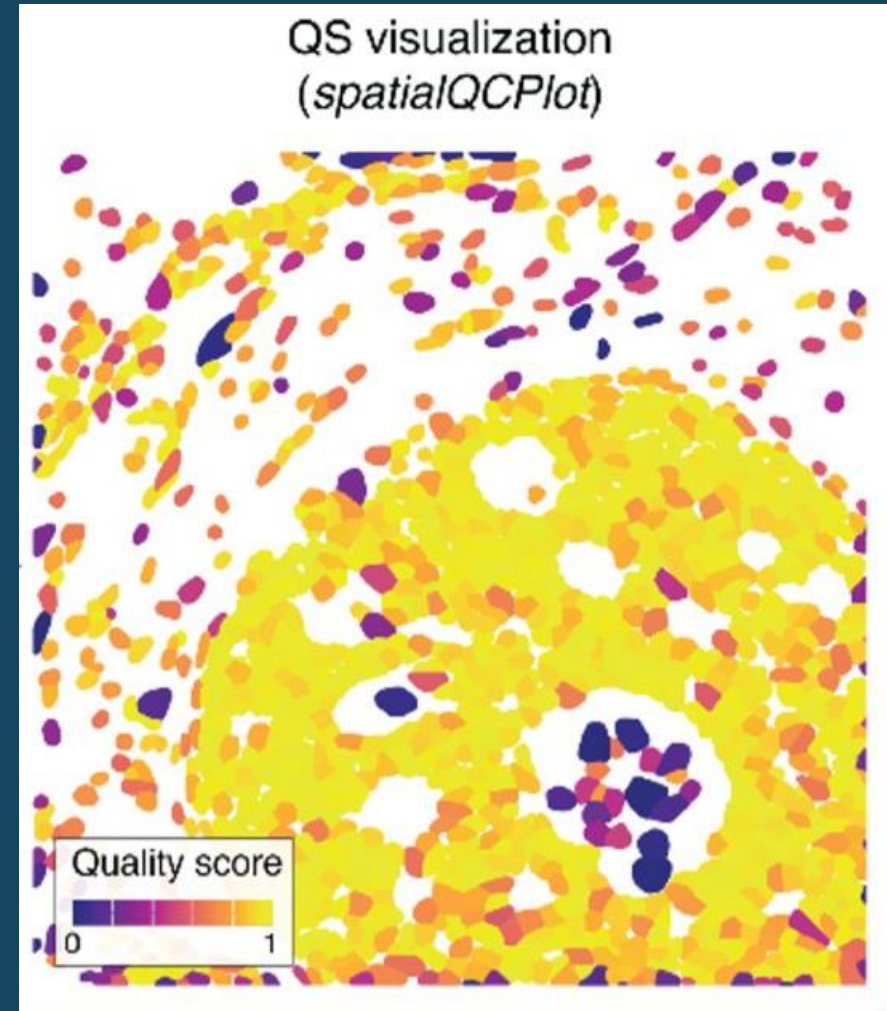
BumpyMatrix molecules assay → ggspavis::plotMolecules()

Beyond: SpatialFeatureExperiment · MoleculeExperiment · TENxIO / SpatialExperimentIO

Righelli, Crowell, Weber, et al, Bioinformatics, 2021

From infrastructure to spatial-aware QC

The object tells us where the cells are.
Now quality control can use that information.



Why spatial-aware quality control?

Imaging-based spatial omics introduces failure modes that are invisible in counts alone.

Low signal

Poor transcript recovery or local signal loss

Morphology

Unexpected cell area or aspect ratio

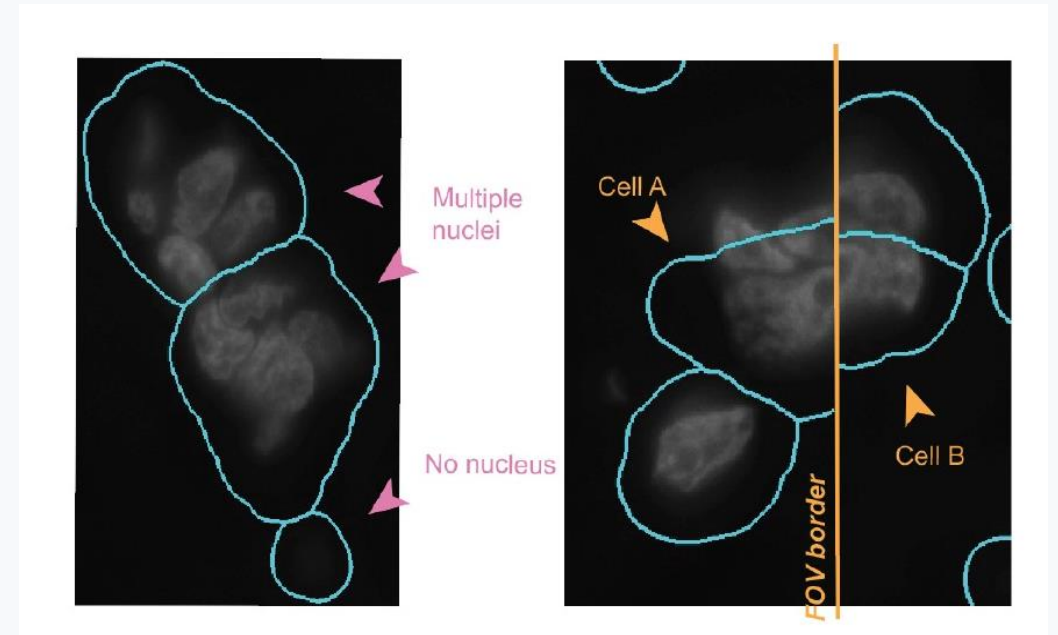
Background

Elevated negative-control signal

Spatial effects

FOV borders and local technical regions

A suspicious cell may be only mildly abnormal on each individual metric — but the pattern becomes informative when metrics and location are considered together.



Segmentation errors and FOV boundaries are spatial phenomena.

Banzi, Righelli, et al, BiorXiv 2025

SpaceTrooper: specialization on top of common infrastructure

Current workflow: SpatialExperiment in, spatial-aware QC out.

Import + standardization

- SpatialExperimentIO readers
- SpaceTrooper update/read helpers
- SpatialExperiment + sf geometries

Exploration in space

- Centroids, FOV maps and polygon-level views
- Global tissue → local field-of-view diagnostics

QC metrics

- Signal density
- Cell area / morphology
- Negative-control ratio
- Distance to border

Integrated quality

- Data-driven regularized model
- Continuous QScore
- Optional low-QScore flags after inspection

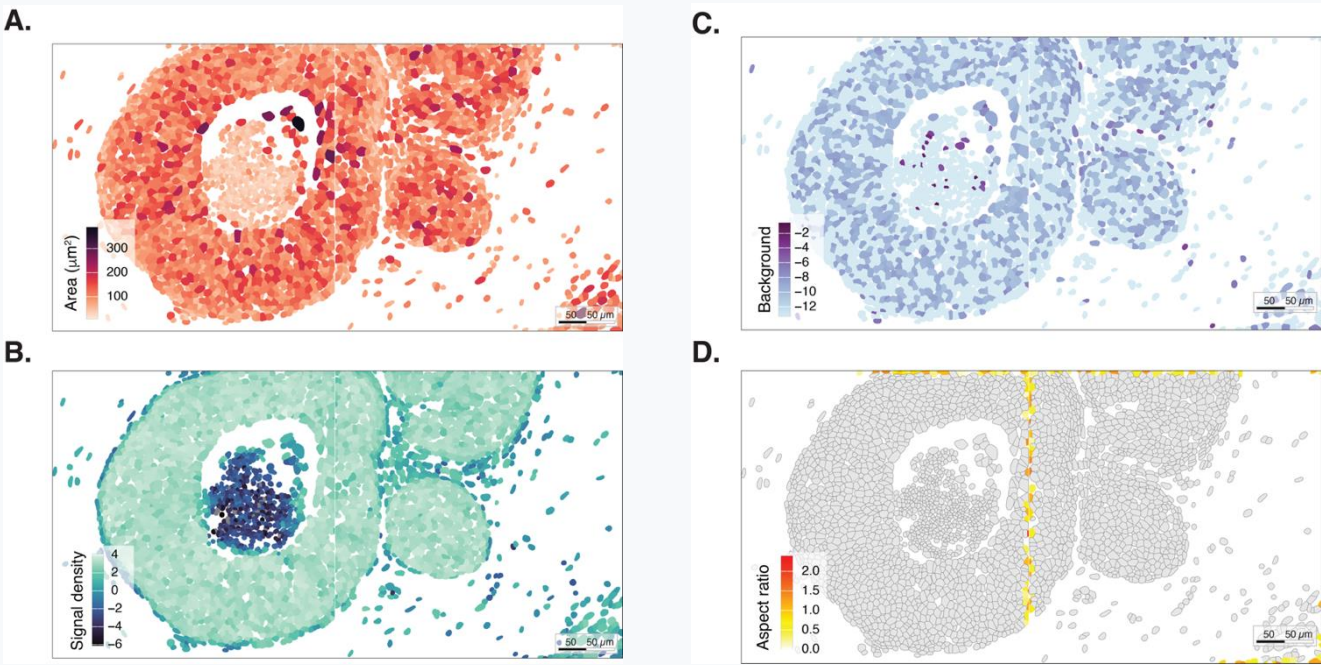
Infrastructure first → technology-aware QC without fragmenting the ecosystem.

Banzi, Righelli, et al, BiorXiv 2025

QC metrics become more informative when mapped back into space

A cell-level metric is not only a number: it has a location and a morphology.

log2SignalDensity	transcripts relative to cell area
Area_um	segmentation / morphology
log2Ctrl_total_ratio	negative controls relative to total signal
log2AspectRatio	shape distortion
dist_border	proximity to a FOV boundary (when accessible)

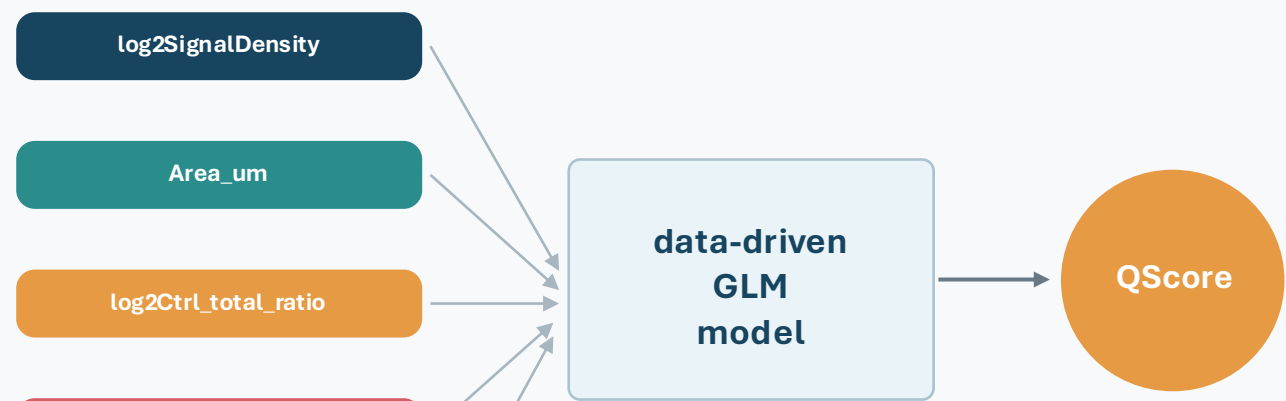


The workshop repeatedly asks: do these metrics identify the same cells, or different technical failure modes?

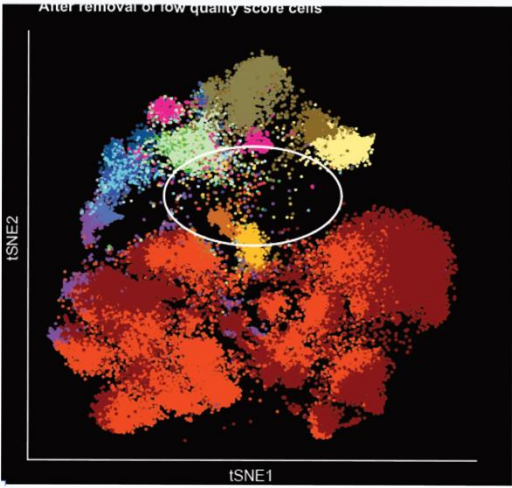
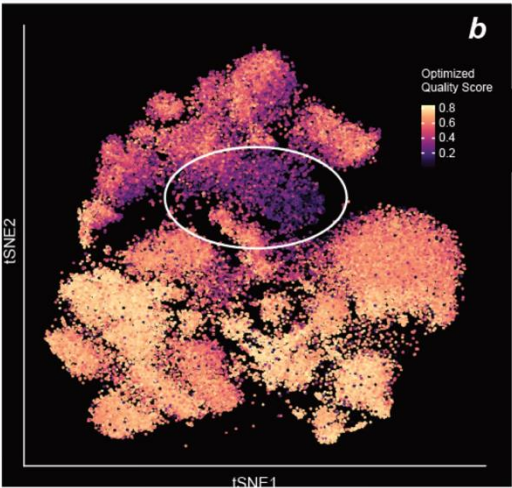
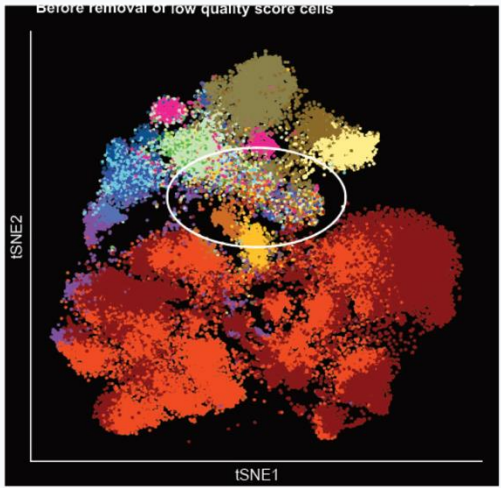
Banzi, Righelli, et al, BiorXiv 2025

From complementary QC signals to QScore

The score integrates weak technical signatures into a continuous per-cell quality measure.



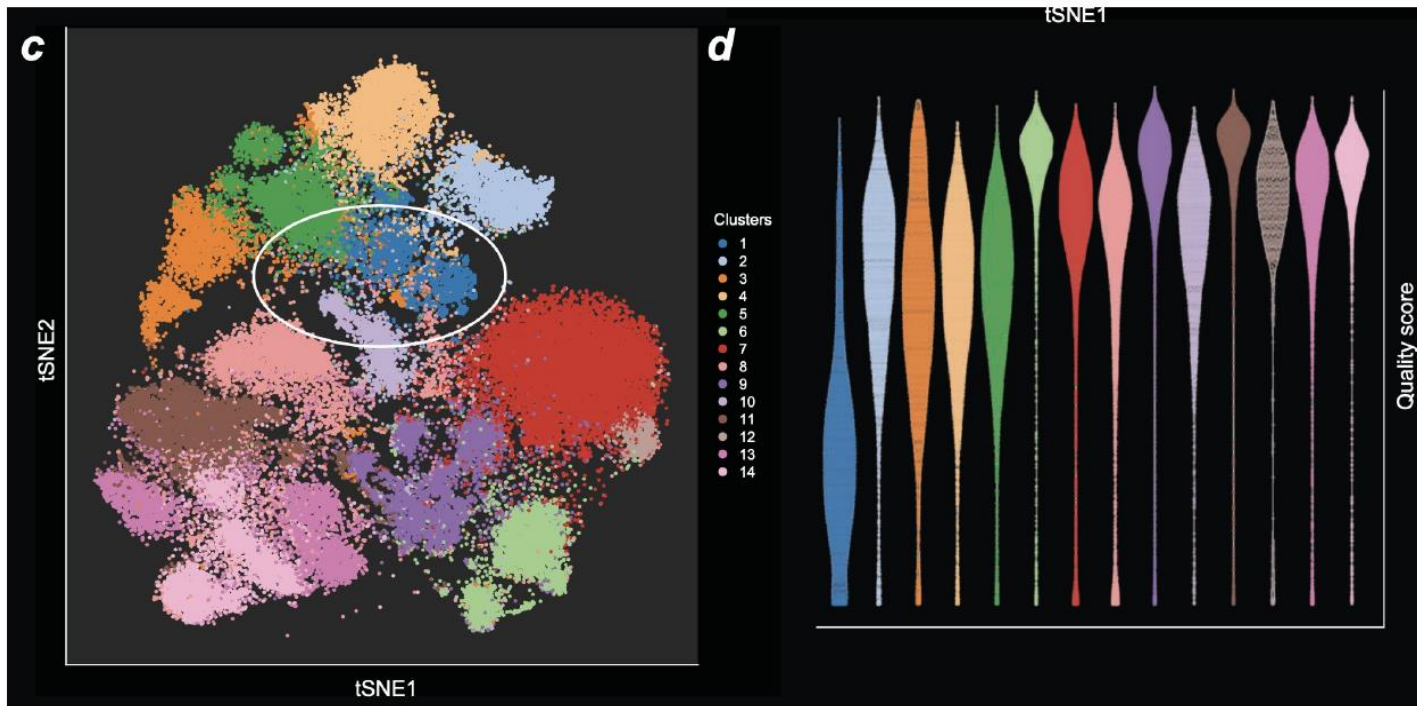
$QScore \in [0,1]$
0 = lower quality · 1 = higher quality



Banzi, Righelli, et al, BiorXiv 2025

Low-quality cells can become a downstream structure

QC decisions can change what clustering and embeddings appear to show.



Do not start from a hard threshold.

Start from metrics + spatial context.

Then decide how filtering affects downstream biology.

Banzi, Righelli, et al, BiorXiv 2025

Live workflow in the workshop

The Quarto carries the code; these are the conceptual hand-offs.



Real data → common representation → technology-aware QC → spatial interpretation

CosMx1k_MouseBrain2: ~950 targets · 48,556 cells · polygons retained · molecule table skipped for the QC practical

Take-home messages

1

Infrastructure matters

Common representations let methods interoperate across platforms.

2

SpatialExperiment stays deliberately small

It represents the experiment without prescribing the analysis.

3

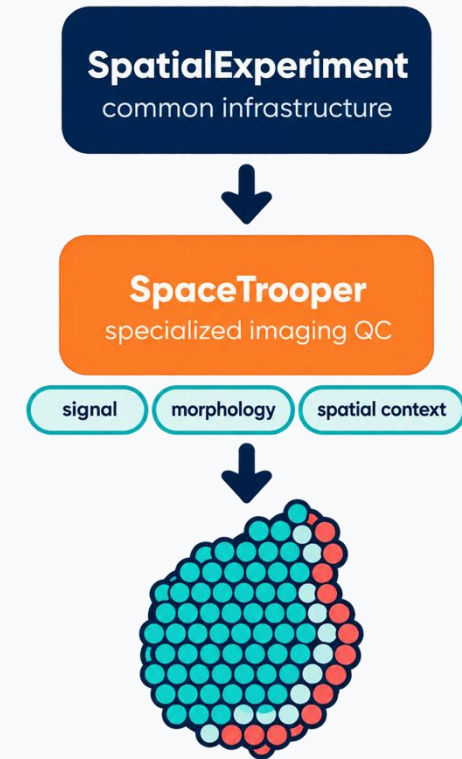
SpaceTrooper adds specialization

Imaging-specific QC combines signal, morphology and spatial effects.

4

Quality must be interpreted in space

QScore is continuous; thresholds are decisions, not biological constants.



Acknowledgements

SpatialExperiment, SpaceTrooper and the Bioconductor community

SpatialExperiment

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SpaceTrooper

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- Oriana Romano
- Mattia Forcato
- Davide Risso
- **Silvio Bicciato**



Bioconductor community

maintainers, contributors, package developers, workshop participants and users